

PROPOSAL

Pension Contribution Processing Automation

Microsoft Power Automate + AI Agent Validation Layer

A practical automation layer that reduces manual load across employer file intake, validation, historical comparison, and pension administration system posting, without rebuilding the existing portal.

Contents

Introduction	03
Context, current state, and the shape of the proposed engagement	
Objectives	03
Outcomes the build will deliver across the 12-week MVP	
Proposed Solution & Flows	04
Two operating flows, mailbox-led and portal-led, sharing one validation layer	
Recommended Architecture	05
Orchestration, agent/API layer, validation services, and audit	
Scope	06
What is included in the 12-week MVP and what sits outside it	
Delivery Plan	07
Indicative 12-week Gantt across discovery, build, UAT, and handover	
UAT Summary	08
Short, rolling validation cycles aligned to milestone outcomes	
Our Delivery Advantage	09
Why this team, this design, and this delivery model	
Commercial Model	10
Two budgetary options, focused build vs. build, manage & operate	
Assumptions, Exclusions & Open Items	11
Conditions the estimate is based on and items to close before signature	
Signature of Approval	12
Sign-off block for client and Newtuple	

01 Introduction

Newtuple proposes a practical automation layer for pension contribution processing. The solution uses **Microsoft Power Automate** where it fits best, mailbox monitoring, attachment handling, notifications, and a controlled **AI agent and validation API** where interpretation, comparison, and auditability need stronger ownership.

The client receives monthly contribution files from multiple employers. These files contain employee contributions, new joiners, exits, retirements, and other membership changes. Today, the work depends on manual review of emails, attachments, prior submissions, validation rules, employer comments, and final upload into the pension administration system.

The proposed engagement focuses on removing this manual load without creating a large workflow platform. The design keeps the existing employer portal intact, introduces an API-accessible agent and validation service, and uses Power Automate for mailbox orchestration and process routing.

This proposal is prepared as a **budgetary estimate**. The final scope, licensing, API behaviour, rule library, and commercial terms should be confirmed during discovery. Estimates carry a $\pm 30\%$ variance until then.

02 Objectives

Outcomes the 12-week MVP will deliver

- Automate intake of employer contribution files from designated Microsoft 365 mailboxes.
- Capture attachments, email context, and employer instructions as part of the submission record.
- Use AI-powered interpretation of employer email commentary for Flow 1, as part of committed scope.
- Retrieve previous-month files from Azure or AWS storage as agreed during discovery.
- Validate files against configurable technical and business rules.
- Compare current submissions with historical files and identify variances.
- Return errors to employers by email for mailbox-led submissions.
- Return errors to the existing portal for portal-led submissions.
- Post validated records to the pension administration system API, where API access is available.
- Maintain a complete audit trail across receipt, validation, communication, resubmission, and upload.

TODAY	AFTER MVP
● Manual review of emails, attachments, and employer commentary ● Cross-checking with prior-month files by hand ● Errors caught late, sometimes after PAS upload ● No central audit trail across receipt to upload	● Mailbox + portal submissions handled through one validation layer ● Historical comparison and variance detection automated ● Errors returned to the right channel before PAS posting ● Full audit ledger across every submission and decision

02 Proposed Solution & Flows

The solution is organized around two operating flows. Both share the same validation and audit layer, but differ in how submissions arrive and how errors are returned to the employer.

FLOW 1

Mailbox, validation, PAS API

Designed for employers who submit by email today

ENTRY

Employer mailbox on Microsoft 365. Submissions arrive as email with attached contribution files.

PATH

Mailbox

 >

Power Automate

 >

AI email agent

 >

Validation core

 >

Historical comparison

 >

PAS API

RESPONSE

Errors and status updates returned to employers by email.

FRS COVERED

FR-001 · FR-002 · FR-003 · FR-004 · FR-005 · FR-006 · FR-007 · FR-008 · FR-010 · FR-011

FR-003 includes AI-powered interpretation of employer email commentary and natural-language instructions, in committed scope for Flow 1.

FLOW 2

Existing portal, validation, portal

Designed for employers using the existing portal

ENTRY

Existing client portal, untouched. Newtuple exposes an API the portal can call for validation.

PATH

Existing portal

 >

Validation core

 >

Historical comparison

 >

Portal response

RESPONSE

Errors and validation results returned synchronously or asynchronously to the existing portal.

FRS COVERED

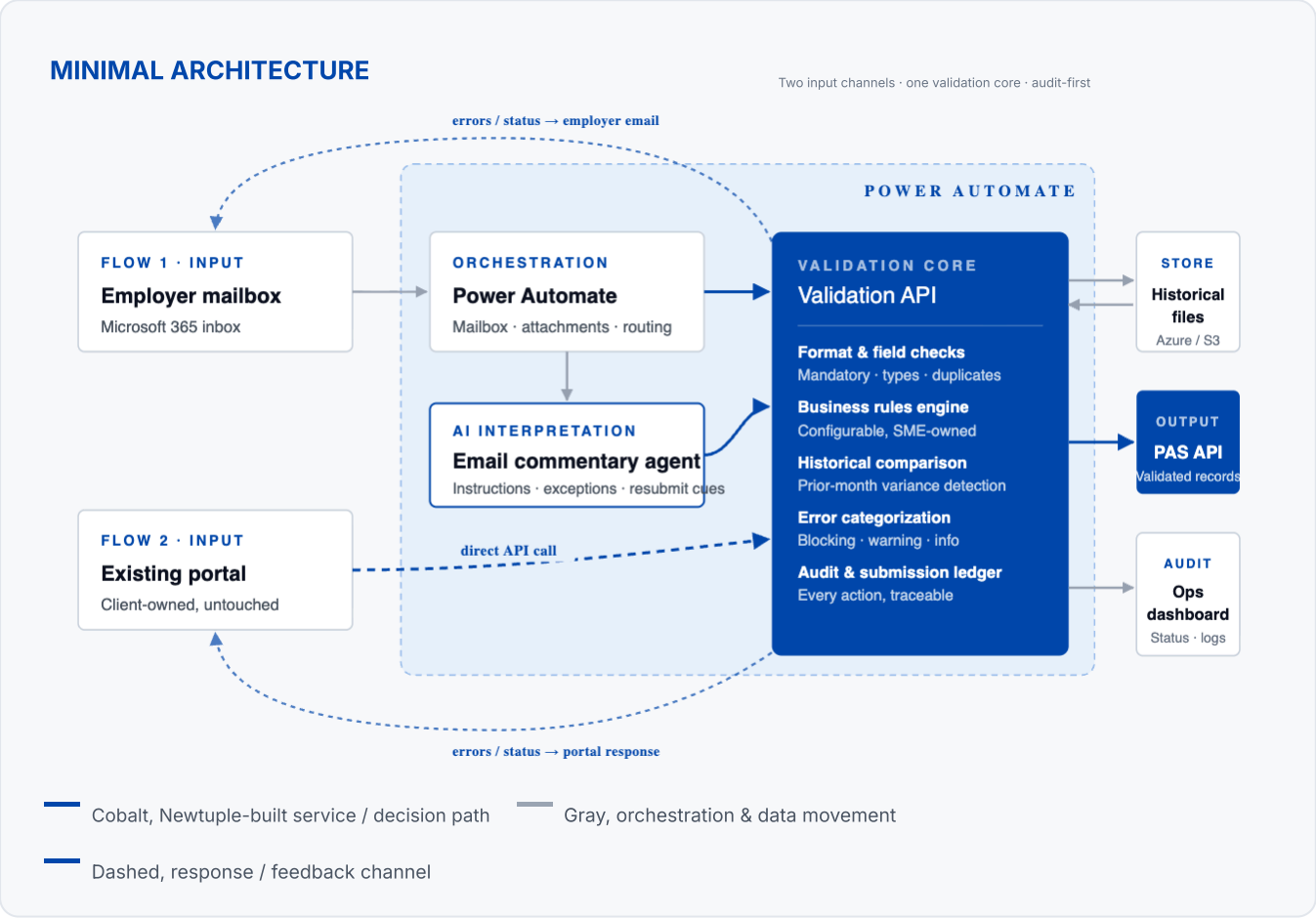
Existing portal · FR-004 · FR-005 · FR-006 · FR-007A · FR-011

The portal remains outside Newtuple build scope. Newtuple provides an API for validation invocation and response handling, no portal rebuild, no UX disruption.

03 Recommended Architecture

Indicative, confirmed in discovery

The architecture separates orchestration, agent interpretation, validation logic, and audit tracking. This keeps Power Automate flows readable while placing business-critical logic in a controlled service layer.



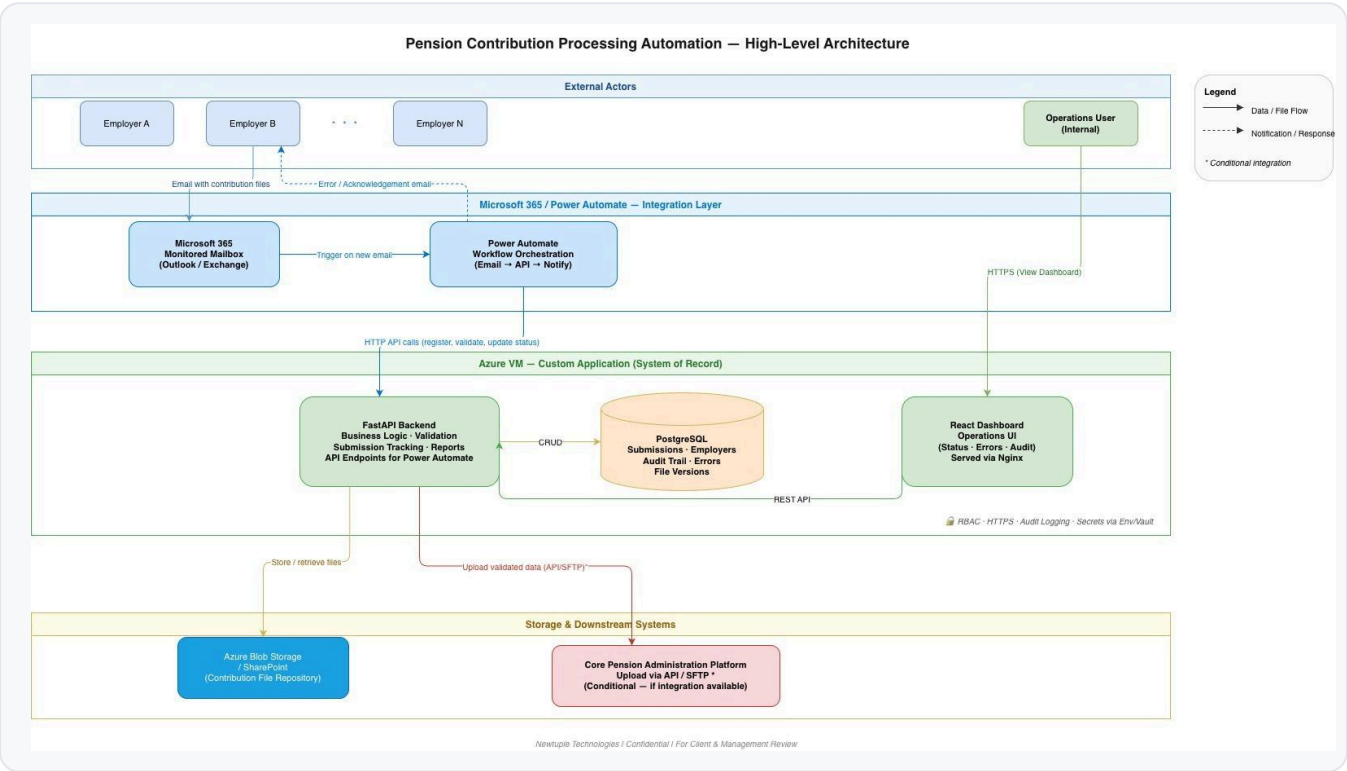
Recommended components

LAYER	RECOMMENDATION
Orchestration	Microsoft Power Automate. Client E5 license is noted, but Premium or Process licensing may still be required for premium connectors.
Agent & API layer	RESTful web API to invoke the agent and validation process synchronously or asynchronously.
AI interpretation	Agent-based processing of employer email commentary and natural-language instructions (Flow 1 only).
Validation engine	Configurable technical and business rule execution with comparison to previous submissions.
Storage & history	Historical files retrieved from the agreed cloud storage location during processing.
PAS integration	Validated data posted to the PAS API, subject to API access, contract, and authentication confirmation.

04 Technical Architecture

System view, indicative

This view shows how the logical flow is realised across Microsoft 365, the Power Automate integration layer, the custom application on Azure, and the downstream pension administration platform. All cross-component access is over HTTPS, with audit logging at every hop.



INTEGRATION LAYER

Microsoft 365 mailbox monitored by Power Automate. Triggers orchestrate intake, call APIs, and emit status notifications back to employers.

SYSTEM OF RECORD

FastAPI backend with business logic and validation.

PostgreSQL holds submissions, employers, audit trail, errors, and file versions. React dashboard for operations.

STORAGE & DOWNSTREAM

Azure Blob or SharePoint as the contribution file repository.

Upload to the pension administration platform via API or SFTP, subject to integration availability.

Security & Operating Controls

- **Authentication** , role-based access across dashboard, validation API, and PAS endpoints.
- **Transport security** , HTTPS everywhere including internal API calls.
- **Audit logging** , every submission, decision, error, notification, and upload recorded with timestamp and actor.
- **Secret management** , credentials held in environment variables or a managed vault, never in source.
- **Hosting** , custom application runs on an Azure VM with Nginx fronting the dashboard and API endpoints.
- **Conditional integration** , PAS posting depends on confirmed API or SFTP availability during discovery.

05

Scope

12-week MVP

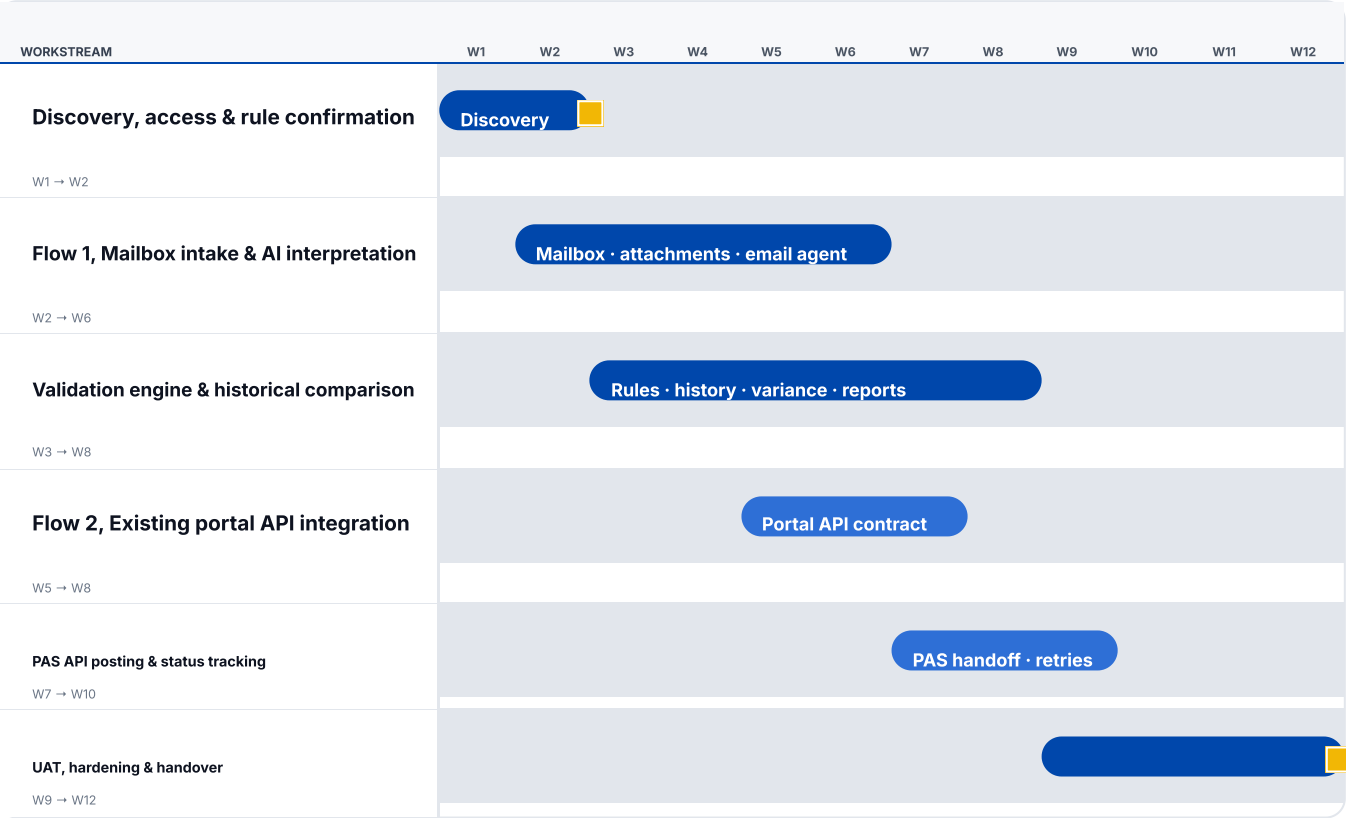
IN SCOPE Capabilities delivered as part of the MVP	
SCOPE AREA	INCLUDED CAPABILITIES
Email processing	Mailbox monitoring, submission detection, attachment extraction, email context capture, and AI interpretation of employer comments.
File management	Central file registration, historical file retrieval, current-to-prior association, version tracking, and resubmission handling.
Validation engine	Format checks, mandatory field checks, data type validation, duplicate detection, contribution amount checks, employee status checks, reconciliation, and configurable business rules.
Error management	Validation report generation, error and warning categorization, automated employer notification for Flow 1, portal-facing error response for Flow 2.
Existing portal integration	API support for the existing portal to call validation services and receive synchronous or asynchronous results.
PAS integration	Validated record posting to the PAS API, upload status capture, and audit logging, subject to API readiness.

OUT OF SCOPE Not included in the 12-week MVP	
● Building or redesigning the existing employer portal. ● Replacing the core pension administration system. ● Third-party security audit, certification, or formal compliance attestation. ● Historical migration beyond data required for validation, testing, and comparison. ● Guaranteed PAS upload automation until PAS API contract, authentication, and payload requirements are confirmed. ● Advanced analytics, employer scorecards, and compliance dashboards unless added as a separate work package.	

Delivery Plan

Indicative 12-week build

The delivery plan is shown as an indicative 12-week build. The sequence may shift after discovery, particularly if licensing, PAS API access, historical storage access, or validation rule confirmation takes longer than expected.



MILESTONE 1 · WEEK 2

Discovery sign-off, requirements, licenses, PAS API contract, storage access, and validation rules confirmed.

MILESTONE 2 · WEEK 8

Validation core stable, Flow 1 & Flow 2 both running against a working rules engine and historical comparison.

GO-LIVE · WEEK 12

MVP ready for controlled production use, with UAT signed off and a handover document delivered.

Plan prioritises a working end-to-end flow early, then adds validation depth, portal integration, PAS posting, and audit hardening.

07 UAT Summary

UAT runs in short cycles across the engagement rather than waiting until the end. This is important because validation behaviour depends on real employer files, historical comparisons, and client-specific business rules.

CYCLE	TIMING	FOCUS
Cycle 1	End of Week 5	Mailbox intake, attachment capture, submission registration, and AI interpretation of email commentary.
Cycle 2	End of Week 8	Validation rules, historical comparison, error reporting, and portal-facing validation response.
Cycle 3	Weeks 10 → 12	PAS posting, retry and status handling, audit trail, operational reports, and handover readiness.

08 Our Delivery Advantage

Newtuple's advantage in this engagement is practical delivery discipline. The proposed design avoids over-engineering, keeps the client's existing portal intact, and puts the highest-risk logic in a service layer that can be tested, audited, and changed without rewriting process flows.

- Clear separation between workflow orchestration and business validation logic.
 - Agent-based interpretation only where it matters, employer comments and instructions in mailbox-led submissions.
 - A reusable validation API that serves both mailbox and portal flows.
- Audit-first processing so every file, decision, error, notification, and upload status can be traced.
 - A delivery model sized for a budgetary decision now, with discovery used to confirm the final build path.
 - A controlled MVP that can expand later into dashboards, employer analytics, exception recommendations, or richer self-service workflows.

WHY NEWTUPLE

Deep GenAI & data engineering practice. Two dozen production-grade AI builds across finance, compliance, and enterprise automation.

OPERATING STYLE

Senior practitioners on every engagement. Weekly demos, sprint discipline, and code in your environment from day one.

AFTER GO-LIVE

A clean handover, or a turnkey operate model under Option B, whichever suits your team's bandwidth.

09

Commercial Model

Indicative, ±30%

The figures below are budgetary estimates based on the current requirement notes and partner clarification. They should be treated as **±30%** until discovery confirms validation complexity, licensing, integration access, PAS API behaviour, and client-side dependencies.

RECOMMENDED

OPTION A

3-month build engagement

USD 36,000 ±30%

Best fit when the client wants a focused MVP build and handover.

Duration · 3 months

Monthly fee · USD 12,000

Model · Build & handover

Post-go-live bug-fix window · 2 weeks

OPTION B

6-month build, manage & operate

USD 5,000 / month

Best fit when the client wants a lower monthly run-rate and ongoing operational support within agreed monthly capacity.

Duration · 6 months

Monthly fee · USD 5,000, paid in advance

Model · Build, manage & operate

Support · Standard IST business hours

What's included in both options

BUILD

Senior delivery team, weekly demos, code in your environment from day one, and a working MVP across both flows.

QUALITY & UAT

Three rolling UAT cycles, defect triage, validated against representative employer files and the agreed rule catalogue.

HANDOVER

Architecture notes, runbook, rule configuration guide, and a walkthrough with the client operations team.

Commercial Notes

- Power Automate Premium or Process licensing may be required even if the client has Microsoft E5.
- Cloud hosting, Microsoft licensing, PAS API charges, storage charges, and third-party service costs are not included unless explicitly agreed.
- Final commercials should be confirmed after discovery validates integration complexity and the final rule catalogue.
- Quote validity should be confirmed before client submission.

Assumptions, Exclusions & Open Items

ASSUMPTIONS

- The client will provide representative employer files, previous-month files, validation rules, and expected outputs during discovery.
- Required access to Microsoft 365, Power Automate, cloud storage, PAS API documentation, and test environments will be available on time.
- The existing portal team will expose the integration point or consume Newtuple's validation API as agreed.
- PAS API posting is subject to confirmed endpoint availability, authentication, payload schema, error handling, and rate limits.
- The client will nominate business users for weekly review, UAT feedback, and rule confirmation.

EXCLUSIONS

- Portal rebuild, new portal UX, or employer identity management changes.
- Large-scale historical data migration or cleansing outside validation test needs.
- Formal security certification, penetration testing, or regulatory attestation.
- Long-term managed service beyond the selected commercial option.
- Enhancements outside the agreed MVP backlog unless approved through change control.

Open Items Before Final Submission

OPEN ITEM	OWNER GROUP	WHY IT MATTERS
Client name and signatory details	Sales	Needed for cover page, signature block, and final quote validity.
PAS API contract and authentication model	Product / client technical team	Determines whether automated upload can be committed in the MVP.
Power Automate licensing position	Client IT / Microsoft admin	May affect implementation cost and process design.
Final validation rule catalogue	Client operations / SME	Controls scope, test cases, and estimate confidence.
Historical storage location and access pattern	Client IT	Required for previous-month comparison and data retrieval timing.
Decision on Option A vs. Option B	Sales / commercial	Needed before presenting a final commercial recommendation.

Commercial Conditions

- Prices are in USD and exclusive of taxes, GST, duties, and statutory charges.
- Invoices are payable within 15 days of invoice date.
- Infrastructure, Power Automate licensing, connector costs, API usage, and third-party software costs are borne by the client.
- Travel, if required, will be billed at actuals.
- Work will be performed during standard IST business hours.
- Scope extensions will be handled through a mutually agreed change request.

11

Signature of Approval

This section will be finalized once the client legal name, authorized signatory, selected commercial option, and quote validity are confirmed.

FOR CLIENT	FOR NEWTUPLE TECHNOLOGIES PRIVATE LIMITED
Name	NameDhiraj Nambiar
Title	TitleFounder
Date	Date

On signature, Newtuple will schedule a discovery kickoff within 5 business days, share the proposed delivery team and weekly cadence, and confirm access requirements across Microsoft 365, Power Automate, cloud storage, and PAS API documentation.